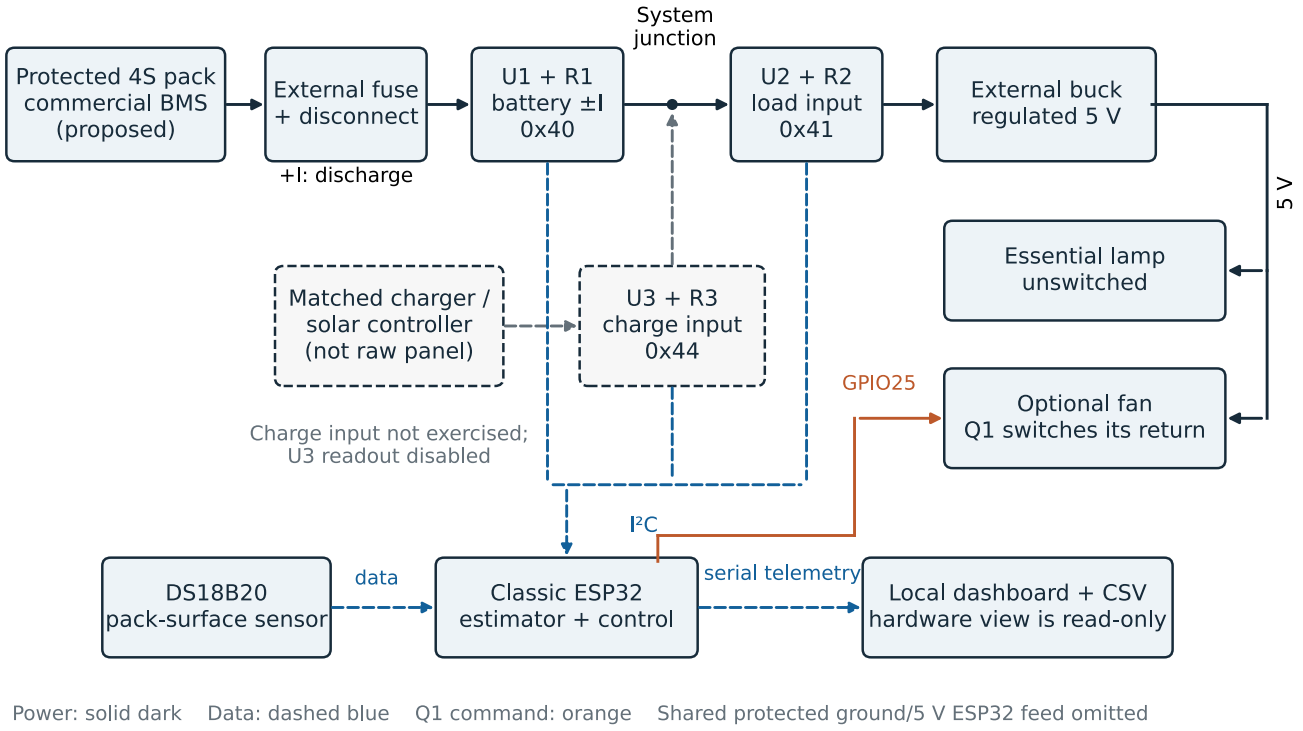


Fig. 1. Proposed Wattsense topology. U1 and U2 monitor signed battery current and load-branch input; U3 provides an unexercised charge-input interface and its readout is disabled in the supplied production configuration. The protected pack, matched charging device, converter and ESP32 are external to the custom board. Power arrows show discharge and charging directions; only the fan return is switched by Q1. The figure omits ground returns, 5 V fuse details and the alternative USB/isolated-data supply arrangements. The principal discharge simulations include battery-powered controller consumption. No assembled hardware is pictured or claimed.

The 100 mm by 90 mm board uses four copper layers, with two inner ground planes. Its nominal stackup is 1.6 mm, with 1 oz outer and 0.5 oz inner copper. Three 0.1 ohm shunts, branch fuses, connectors, a fan MOSFET and flyback diode provide the interface around purchased pack and converter modules. The declared design limits are 2 A on the battery path and 1.5 A continuous on the 5 V path, including controller demand. These are design limits pending qualification, not measured current ratings. Four layers alone do not establish thermal or electromagnetic performance.

An AO3400A switches the fan return; the lamp remains connected. The startup model uses 48 milliohms for MOSFET conduction, consistent with the selected part's room-temperature maximum at 2.5 V gate drive [6]. Fuse cold-resistance inputs are nominal data-sheet values [7], not predictions of hot resistance or clearing time. A DS18B20 is intended to monitor pack-surface temperature. The ESP32 makes local decisions and exports telemetry to a read-only dashboard. The switch state in telemetry is a command, not an independent measurement of load interruption.

A. Charge and runtime estimates

Let s be estimated state of charge in percent, C the configured capacity in Ah before reserving charge, I the signed battery current in A and dt the elapsed time in seconds. The implemented update is

$$s[k+1] = clip(s[k] - 100\, I_{eff}\, dt / (3600\, C),\ 0,\ 100). \quad (1)$$

Here I_{eff} equals I during discharge and 0.98 I during charging. The nominal voltage V_{nom} is 12.8 V and reserve s_{res} is 20%. Estimated energy above reserve and the load-based trigger time are

$$E_{est} = C\, V_{nom}\, max(s - s_{res},\ 0) / 100, \quad (2)$$

$$T_{est} = E_{est} / P_{U2},\ P_{U2} > 0.2\ W. \quad (3)$$

E_{est} is in Wh and T_{est} in hours. P_{U2} is measured upstream of the load shunt and converter, so it includes downstream conversion and distribution losses. It excludes battery/internal and some upstream losses. Equation (2) also substitutes nominal voltage for an integrated voltage curve. Consequently, excluding future charging does not make (3) a minimum-runtime guarantee. A separate net-discharge display uses battery terminal power with a 3 s exponential filter; that filtered denominator is not the denominator of (3).

B. Decision sequence and invalid data

Figure 2 summarizes the local policy. With the hardware armed and measurements valid, three consecutive seconds below T_{min} = 3 h command the fan off and latch the decision. Restoration requires an explicit request and a prospective estimate of at least T_{min} + 0.25 h. The reference firmware configuration reserves a provisional 3.5 W increment for that check; this must be replaced by measured optional-load input power. The principal discharge comparison starts with the fan on and does not exercise restoration.